

Prof. Dr. Lincy Pyl and Selection Committee
Department of Mechanics of Materials and Constructions (MeMC)
Vrije Universiteit Brussel

Subject: Motivation letter – PhD candidate, Mechanics of Materials and Constructions (M²S)
Introduction video: <https://drive.google.com/file/d/14s-V07z3fopEllnUP8vmMJPMxQW57mg9/view>

Dear Prof. Pyl and Selection Committee,

I am Pooja Awasare, a civil and environmental engineer, and I am applying for the PhD position in Mechanics of Materials and Constructions at VUB. Materials and structures have stayed with me since my civil engineering studies, and I want to take that interest into doctoral research with your group at MeMC.

During my Diploma and Bachelor's in Civil Engineering in India I spent many hours in the materials laboratory with concrete, steel, soil, aggregates and water. Those practical courses shaped the way I look at engineering problems. From those labs I learned to stay with a result and keep checking until I understood what had caused it, and that same way of working is still with me when I read, test or plan something carefully.

I later completed a Master's in Environmental Engineering at HTW Dresden. My thesis, *Agile Project Management in the Construction Industry*, looked at how construction teams can manage projects and use resources more carefully. While writing it I kept coming back to materials and structures, because they sit underneath almost every construction decision. That brought me back to the laboratory side of civil engineering and to questions about how materials behave and how structures use them in sustainable practice.

In Belgium I worked with a construction partner on circular practices in day-to-day project delivery. The work was practical and close to site and process questions around waste, reuse and material flows. It strengthened my wish to go deeper into the mechanical behaviour of materials and structures through laboratory testing and careful analysis inside a research group.

I have been following MeMC and reading about your group's work over the past months. I read about Prof. Pyl's research on lightweight structures and on the mechanical performance of materials, where experiments are linked with analytical and numerical models. I also followed MeMC's work on resource-efficient and circular building systems, including the wider programme around innovative structures that carry load well while using materials carefully and staying open to reuse. Within M²S I also read about research on damage and monitoring of construction materials and structures, and that line of work stayed with me because it sits so close to the laboratory habits I built during my civil engineering studies. Reading those pages and papers helped me understand how your group connects laboratory work, modelling and real construction questions.

When I read the vacancy carefully, I could see the shape of the role quite clearly: a full four-year PhD within M²S, research that grows from advanced characterisation and monitoring as well as mechanics-driven design of sustainable materials and structures, presenting and publishing the work, and a limited share of teaching through exercise and laboratory sessions at the Elsen campus. What I find especially interesting is that the PhD project itself is meant to be developed together with the supervisor from my own strengths and research interests, and that feels amazing to me, because it means the topic can grow honestly from what I already care about and from the direction of the group.

I am based in Pune at the moment, with earlier years of study and work in Germany and Belgium behind me. English is the language I work in most comfortably, and I have only basic Dutch from my time in Belgium. I am willing to learn Dutch further, and of course I will keep working on it so I can take part more fully in department life and in the educational activities that are taught in Dutch. I would like to join MeMC for a full PhD path and grow through the experimental and analytical work of the group.

Thank you for considering my application and for watching my introduction video. I look forward to the possibility of speaking with you.

Yours sincerely,

Pooja Awasare